

Regularization and Variable Selection

Instructor: Mohammad Arashi (FUM, UP)



دانشگاه فردوسی مشهد
Ferdowsi University of Mashhad



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

- 1 Regularization and Penalization
- 2 Penalized estimators

Consider the multiple regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (1.1)$$

where $\mathbf{y}$ is an $n \times 1$ vector of responses, $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$ is an $n \times p$ fixed-design matrix including the covariates $\mathbf{x}_i \in \mathbb{R}^p$, $i = 1, \dots, n$, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p) \in \mathbb{R}^p$ is the regression coefficients' vector, and $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of measurement errors (we eliminated the intercept from our model) with $E(\boldsymbol{\epsilon}) = \mathbf{0}$.

If the columns of the design matrix $\mathbf{X}$ are linearly independent, then the estimator of $\boldsymbol{\beta}$ is the unique solution $\mathbf{b} = \hat{\boldsymbol{\beta}}$ of the LS problem (see Session 1)

$$\min \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2 \quad \text{subject to} \quad \mathbf{b} \in \mathbb{R}^p, \quad (1.2)$$

where for $\mathbf{a} = (a_1, \dots, a_n)^\top$, $\|\mathbf{a}\|_p = (\sum_{i=1}^n a_i^p)^{\frac{1}{p}}$.

Recall from Session 1, if we have a high-dimensional case, or rank deficient design matrix, or a multicollinear situation, then, we need to take an action for the OLS estimate.

Let's focus on not-independent columns case: I elaborate more.

Under situations in which the matrix $\mathbf{X}^\top \mathbf{X}$ is ill-conditioned due to near/exact linear relationship among the regressors of $\mathbf{X}$, we always find a linear combination of the columns in $\mathbf{X}$ which is exactly equal to one other (not full rank), and one may have $\mathbf{X}\beta = \mathbf{X}(\beta + \zeta)$ for every ζ in the null space of $\mathbf{X}$. It is then impossible to separate the contribution of the individual covariates. The uncertainty with respect to the covariate responsible for the variation explained in $\mathbf{y}$ is often reflected in the fit of the linear regression model to data by a large error of the estimates of the regression parameters corresponding to the collinear covariates and, consequently, usually result large values of the estimates.

Consequently, without applying constraint, it is impossible to infer or estimate β from data.

It is not hard to decide which constraint. We want to estimate β , thus make the constraint on β . Intuitively, you may think that the magnitude of β causes the problem (since $\mathbf{X}\beta = \mathbf{X}(\beta + \zeta)$). Thus, make a constraint based on the magnitude of β and since it is a vector, use a norm. Therefore, the penalized regression problem has form

Constrained form

$$\min \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2 \quad \text{subject to} \quad \|\beta\| \leq c, \quad c > 0. \quad (1.3)$$

Extremely important: n is the number of observations in the training set, if we have enough sample points to divide into training and test sets.

Penalized least squares estimator

Equivalently, using the Lagrangian multipliers, the solution to the constrained LS problem is given by (equivalently)

Lagrange multiplier form

$$\hat{\beta}_{\lambda}^{\text{Pen}} = \arg \min_{\beta \in \mathbb{R}^p} \{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\| \} \quad (1.4)$$

You like this one ↓ more!!!

$$\hat{\beta}_{\lambda}^{\text{Pen}} = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \sum_{i=1}^p p_{\lambda}(|\beta_j|) \right\} \quad (1.5)$$

which we use $p_{\lambda}(|\beta_j|) = \lambda p(|\beta_j|)$ (often).

Relation between c and λ

What is the relation/connection between the upper bound condition parameter c and the tuning (shrinkage) parameter λ ?

Consider $p(x) = x$. Then, $\sum_{i=1}^p |\beta_i| \leq c$ shrinks the coefficients and as c gets smaller the amount of shrinkage increases. For the case $c \rightarrow 0$, $\beta_i \rightarrow 0$, $\forall i = 1, \dots, p$. In contrast, As λ increases, then $\sum_{i=1}^p |\beta_i|$ decreases, i.e. the coefficients become shrunken towards zero. Thus, c and λ have reverse (contrary) performance.

Now the questions are: Which penalty function? How to select λ ?

Good penalty

Before we consider some well-known and useful penalty functions for the question “Which penalty function?”, we define a good penalty function. According to Fan and Li (2001)¹, any good penalty function must have the following properties:

- Unbiasedness
- Sparsity
- Continuity

Thus, our target is to focus on penalty functions with good properties. However, it is not always achievable and sometimes we use a penalty that fulfills only one property (e.g. continuity in the L_2 -norm case as in the ridge penalty).

¹Fan, J. and Li, R. (2001) Variable selection via nonconcave penalized likelihood and its oracle properties, J. Amer. Statist. Assoc., 96(456), 1348-1360

Good penalty cont.

- Unbiasedness** The resulting estimator is nearly unbiased when the true unknown parameter is large to avoid unnecessary modeling bias. Thus, we need $p_\lambda(\cdot)$ that results in an estimator that is nearly unbiased for large components of β .
- Sparsity** The resulting estimator is a thresholding rule, which automatically sets small estimated coefficients to zero to reduce model complexity. Thus, we need $p_\lambda(\cdot)$ that results in an estimator that sets some components of β to 0.
- Continuity** The resulting estimator is continuous in $(\mathbf{X}^\top \mathbf{y})$ data in the training set to avoid instability in model prediction.

Good penalty cont.

In order to digest the good properties and a way to check, we consider the orthonormal design case in which $\mathbf{X}^\top \mathbf{X} = \mathbf{X} \mathbf{X}^\top = \mathbf{I}_p$. Then,

$$\begin{aligned}
 \|\mathbf{y} - \mathbf{X}\beta\|_2^2 &= \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta \\
 &= \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta + 2\mathbf{y}^\top \mathbf{X} \mathbf{X}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X} \mathbf{X}^\top \mathbf{y} \\
 &= \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X} \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{X} \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{X} \mathbf{X}^\top \mathbf{y} \\
 &\quad - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta \\
 &= \mathbf{y}^\top (\mathbf{I} - \mathbf{X} \mathbf{X}^\top)^\top (\mathbf{I} - \mathbf{X} \mathbf{X}^\top) \mathbf{y} + (\mathbf{X}^\top \mathbf{y} - \beta)^\top (\mathbf{X}^\top \mathbf{y} - \beta) \\
 &= \|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \|\hat{\beta} - \beta\|_2^2.
 \end{aligned} \tag{1.6}$$

Note here, $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{y}$ and $\hat{\mathbf{y}} = \mathbf{X} \hat{\beta} = \mathbf{X} \mathbf{X}^\top \mathbf{y}$.

Let's take $\mathbf{z} = \mathbf{X}^\top \mathbf{y} = (z_1, \dots, z_p)$.

Good penalty cont.

Thus, minimizing $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \sum_{j=1}^p p_\lambda(|\beta_j|)$ is equivalent to minimizing

$$\frac{1}{2}(z_j - \beta_j)^2 + p_\lambda(|\beta_j|) \quad (1.7)$$

elementwise for each $\beta_j, j = 1, \dots, p$.

First note sometimes to avoid unnecessary coefficient factor, we use $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$, or $\frac{1}{n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$, or $\frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$, instead. Further, in sequel, I ignore the subscript j and use z and β instead of z_j and β_j , for the ease of understanding

To minimize the new objective function, we get derivatives. Thus, the first order derivative of (1.7) is (ignoring j)

$$\begin{aligned} & -z + \beta + p'_\lambda(|\beta|) \times \text{sgn}(\beta) \\ = & \text{sgn}(\beta) (|\beta| + p'_\lambda(|\beta|)) - z \\ = & q(\beta) \quad (\text{say}) \end{aligned} \quad (1.8)$$

Then we seek for β such that $q(\beta) = 0$.

Good penalty cont.

Unbiasedness Since the OLS estimator (z , here) is unbiased, we require for large values $|\beta|$, our estimator becomes equal to z . From (1.8), it is easy to see that when $p'_\lambda(|\beta|) = 0$, for large $|\beta|$, then the resulting estimator is z ($\beta = z$). Thus the condition for unbiasedness is

$$p'_\lambda(|\beta|) \rightarrow 0 \quad \text{as} \quad |\beta| \rightarrow \infty. \quad (1.9)$$

Good penalty cont.

Sparsity In this case $q(0) = 0$. It means we have $\beta = 0$ that minimizes (1.7). This satisfies if $q(\beta)$ changes sign as β does at the origin.

$$\begin{aligned} q(\beta) > 0 &\implies z < \operatorname{sgn}(\beta) (|\beta| + p'_\lambda(|\beta|)) \\ &\stackrel{\beta > 0}{\implies} z < \inf_{\beta} (|\beta| + p'_\lambda(|\beta|)) \end{aligned}$$

$$\begin{aligned} q(\beta) < 0 &\implies z > \operatorname{sgn}(\beta) (|\beta| + p'_\lambda(|\beta|)) \\ &\stackrel{\beta < 0}{\implies} -z < \inf_{\beta} (|\beta| + p'_\lambda(|\beta|)) \end{aligned}$$

Thus we need $|z| < \inf_{\beta} (|\beta| + p'_\lambda(|\beta|))$. This condition is satisfied if

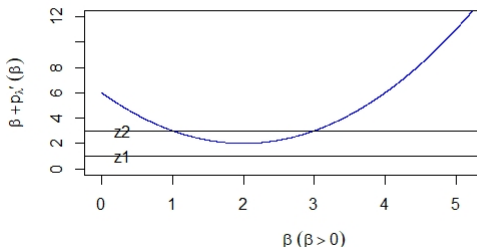
$$\inf_{\beta} (|\beta| + p'_\lambda(|\beta|)) > 0 \quad (1.10)$$

since $|z| > 0$ always.

Good penalty cont.

Continuity Continuity means if we change the input data, we must not have jump. For $|z| < \inf_{\beta} (|\beta| + p'_{\lambda}(|\beta|))$, β is shrunk to zero. For $|z| > \inf_{\beta} (|\beta| + p'_{\lambda}(|\beta|))$ ² two crossings may exist as shown in the this figure; the larger one is a penalized least squares estimator. This implies $\beta = 0$ must be minimizer of $(|\beta| + p'_{\lambda}(|\beta|))$. Thus

$$\arg \inf_{\beta} (|\beta| + p'_{\lambda}(|\beta|)) = 0 \quad (1.11)$$



²Note here, the training data z is changed

Ridge regression

As you are familiar (from Session 1), the ridge regression comes first. Using the L_2 -norm, take $p(|\beta_j|) = \beta_j^2$. Thus, we solve the following constraint problem

$$\min \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_2^2 \leq c, \quad c > 0. \quad (2.1)$$

where $\|\boldsymbol{\beta}\|_2^2 = \boldsymbol{\beta}^\top \boldsymbol{\beta} = \sum_{i=1}^p \beta_i^2$. Thus,

$$\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ PS_{\lambda}^{\text{ridge}}(\boldsymbol{\beta}) \right\}, \quad (2.2)$$

where the penalized least squares criterion is

$$PS_{\lambda}^{\text{ridge}}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2. \quad (2.3)$$

Ridge regression cont.

From (for both low and high dimensional cases, you know from Session 1)

$$\frac{\partial PS_{\lambda}^{\text{ridge}}(\beta)}{\partial \beta} = \frac{\partial}{\partial \beta} \mathbf{y}^{\top} \mathbf{y} - 2\beta^{\top} \mathbf{X}^{\top} \mathbf{y} + \beta^{\top} \mathbf{X}^{\top} \mathbf{X} \beta + \lambda \beta^{\top} \beta = -2\mathbf{X}^{\top} \beta + 2\mathbf{X}^{\top} \mathbf{X} \beta + 2\lambda \beta = \mathbf{0} \quad (2.4)$$

the ridge estimator has form

$$\hat{\beta}^{\text{ridge}}(\lambda) = (\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^{\top} \mathbf{y}. \quad (2.5)$$

If $\mathbf{X}^{\top} \mathbf{X}$ is invertible, using the OLS estimator $\hat{\beta}$, we have

$$\begin{aligned} \hat{\beta}^{\text{ridge}}(\lambda) &= (\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}_p)^{-1} (\mathbf{X}^{\top} \mathbf{X}) (\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{X}^{\top} \mathbf{y} \\ &= \mathbf{C}(\lambda)^{-1} \hat{\beta}, \quad \mathbf{C}(\lambda) = (\mathbf{I}_p + \lambda (\mathbf{X}^{\top} \mathbf{X})^{-1}). \end{aligned} \quad (2.6)$$

Ridge regression cont.

Augmented data construction: what is the benefit? In any context the OLS is valid, you can get the ridge using augmentation.

Consider the following augmented data

$$\mathbf{X}^* = \begin{bmatrix} \mathbf{X} \\ \sqrt{k}\mathbf{I}_p \end{bmatrix}, \quad \mathbf{Y}^* = \begin{bmatrix} \mathbf{Y} \\ \mathbf{0} \end{bmatrix}. \quad (2.7)$$

Assume the following multiple regression model,

$$\mathbf{Y}^* = \mathbf{X}^* \boldsymbol{\beta} + \boldsymbol{\epsilon}^*, \quad (2.8)$$

where $\boldsymbol{\epsilon}^*$ is an $(n + p)$ -vector of i.i.d. random variables. Then, the OLS estimator of $\boldsymbol{\beta}$ is obtained as

$$\begin{aligned} \boldsymbol{\beta}_n^* &= \min_{\boldsymbol{\beta}} \left\{ (\mathbf{Y}^* - \mathbf{X}^* \boldsymbol{\beta})^\top (\mathbf{Y}^* - \mathbf{X}^* \boldsymbol{\beta}) \right\} \\ &= (\mathbf{X}^{*\top} \mathbf{X}^*)^{-1} \mathbf{X}^{*\top} \mathbf{Y}^* \\ &= (\mathbf{X}^\top \mathbf{X} + k\mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{Y} = \hat{\boldsymbol{\beta}}^{\text{ridge}}(k). \end{aligned}$$

Thus, the OLS estimator of the augmented data is indeed the RRE of the normal data.

Ridge regression cont.

There always exists a $\lambda > 0$ such that the risk of $\hat{\beta}^{\text{ridge}}(\lambda)$ is lesser than that of the OLS estimator. How should we find (estimate) λ then?

(Saleh et al., 2019) Define the generalized cross validation (GCV) as the weighted average of predicted square errors

$$\text{Var}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\mathbf{x}_i^\top \hat{\beta}^{\text{ridge}}(\lambda) - y_i \right)^2 w_i(\lambda), \quad (2.9)$$

where

$$w_i(\lambda) = \frac{(1 - a_{ii}(\lambda))}{1 - \frac{1}{n} \text{tr} \mathbf{A}(\lambda)}, \quad \mathbf{A}(\lambda) = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top, \quad (2.10)$$

where $a_{ii}(\lambda)$ is the i th diagonal element of $\mathbf{A}(\lambda)$.

Ridge regression cont.

A computationally efficient version of the GCV function is

$$\text{GCV}(\lambda) = \frac{1}{n} \frac{\|(\mathbf{I}_n - \mathbf{A}(\lambda))\mathbf{y}\|_2^2}{\left[\frac{1}{n} \text{tr}(\mathbf{I}_n - \mathbf{A}(\lambda))\right]^2} \quad (2.11)$$

The GCV estimator of λ is then given by

$$\lambda_{\text{opt}} = \arg \min_{\lambda > 0} \text{GCV}(\lambda).$$

Ridge regression cont.

Ridge is important!!!

Package	Description
lmridge	A comprehensive R package for ridge regression
lrmest	Computes estimators such as OLS, ordinary RR, Liu
ltsbase	Computes ridge and Liu estimates based on the least trimmed squares method
penalized	Computes LASSO, ridge and fused LASSO
glmnet	Computes LASSO, ridge, and elastic-net (we used in Session 1)
ridge	Computes ridge for linear and logistic models for genome-wide SNP data
RXshrink	Computes generalized ridge estimator using the ML method
rrBLUP	Computes ridge estimator for genomic selection
RidgeFusion	Computes ridge estimator for fusion statistical learning, QDA, and cluster analysis

LASSO

Major drawbacks of the ridge:

(for the orthonormal design, since $p_\lambda(|\beta|) = \lambda\beta^2$)

$$p'_\lambda(|\beta|) = 2\lambda\beta \rightarrow \infty \quad \text{as } \beta \rightarrow \infty \implies \text{biased}$$

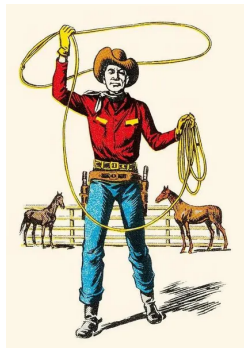
$$\inf_{\beta} (|\beta| + 2\lambda\beta) = 0 \implies \text{not sparse}$$

$$\arg \inf_{\beta} (|\beta| + 2\lambda\beta) = 0 \implies \text{continues}$$

no feature selection; estimate all the p features, thus no parsimony; in high-dimension, does not give an interpretable model.

Tibshirani (1996) proposed the least absolute shrinkage and selection operator (LASSO).

Tibshirani, R. (1996) Regression shrinkage and selection via the lasso, J. Royal Statist. Soc. Ser. B, 58(1), 267-288.



LASSO

Using the L_1 -norm, take $p(|\beta_j|) = |\beta_j|$. Thus, we solve the following constraint problem

$$\min \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \leq c, \quad c > 0, \quad (2.12)$$

where $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|$. Thus,

$$\hat{\boldsymbol{\beta}}^{\text{LASSO}}(\lambda) = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \{PS_{\lambda}^{\text{LASSO}}(\boldsymbol{\beta})\}, \quad (2.13)$$

where the penalized least squares criterion is

$$PS_{\lambda}^{\text{LASSO}}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1. \quad (2.14)$$

Note $\|\boldsymbol{\beta}\|_1 = \mathbf{1}_p^{\top} |\boldsymbol{\beta}|$, where $\mathbf{1}_p$ is a p -tuple of 1's and $|\boldsymbol{\beta}| = (|\beta_1|, \dots, |\beta_p|)^{\top}$.

LASSO cont.

There is no closed form solution for (2.13). Now, consider the orthonormal design and take $PS_{\lambda}^{\text{LASSO}}(\beta) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda\|\beta\|_1$.

Differentiating $PS_{\lambda}^{\text{LASSO}}(\beta)$ and setting the resultant to zero, we obtain ($\mathbf{X}^{\top}\mathbf{X} = \mathbf{I}_p$)

$$-\mathbf{X}^{\top}\mathbf{y} + \beta + \lambda\text{sgn}(\beta) = \mathbf{0} \implies -\hat{\beta} + \beta + \lambda\text{sgn}(\beta) = \mathbf{0}, \quad (2.15)$$

where $\text{sgn}(\beta) = (\text{sgn}(\beta_1), \dots, \text{sgn}(\beta_p))^{\top}$. Now, the j th marginal component of (2.15) is given by $\hat{\beta}_j^{\text{LASSO}}(\lambda)$,

$$-\hat{\beta}_j + \hat{\beta}_j^{\text{LASSO}}(\lambda) + \lambda\text{sgn}\left(\hat{\beta}_j^{\text{LASSO}}(\lambda)\right) = 0; j = 1, 2, \dots, p. \quad (2.16)$$

LASSO cont.

Now, we have three cases:

(i) $\text{sgn}\left(\hat{\beta}_j^{\text{LASSO}}(\lambda)\right) = +1$, then (2.16) reduces to

$$-\hat{\beta}_j + \hat{\beta}_j^{\text{LASSO}}(\lambda) + \lambda = 0, \quad j = 1, 2, \dots, p. \quad (2.17)$$

Hence,

$$0 < \hat{\beta}_j^{\text{LASSO}}(\lambda) = \left(\hat{\beta}_j - \lambda\right) = \left|\hat{\beta}_j\right| - \lambda, \quad (2.18)$$

with, clearly, $\hat{\beta}_j > 0$ and $\left|\hat{\beta}_j\right| > \lambda$.

LASSO cont.

(ii) $\text{sgn} \left(\hat{\beta}_j^{\text{LASSO}}(\lambda) \right) = -1$, then we have

$$-\hat{\beta}_j + \hat{\beta}_j^{\text{LASSO}}(\lambda) - \lambda = 0; \quad j = 1, 2, \dots, p. \quad (2.19)$$

Hence,

$$\begin{aligned} 0 > \hat{\beta}_j^{\text{LASSO}}(\lambda) &= \hat{\beta}_j + \lambda = -|\hat{\beta}_j| + \lambda \\ \hat{\beta}_j^{\text{LASSO}}(\lambda) &= -(|\hat{\beta}_j| - \lambda) \end{aligned} \quad (2.20)$$

with, clearly, $\hat{\beta}_j < 0$ and $|\hat{\beta}_j| > \lambda$.

LASSO cont.

- (iii) For $\hat{\beta}_j^{\text{LASSO}}(\lambda) = 0$, we have $-\hat{\beta}_j + \lambda\gamma = 0$ for some $\gamma \in [-1, 1]$. Hence, we obtain $\hat{\beta}_j = \lambda\gamma$, which implies $|\hat{\beta}_j| \leq \lambda$.

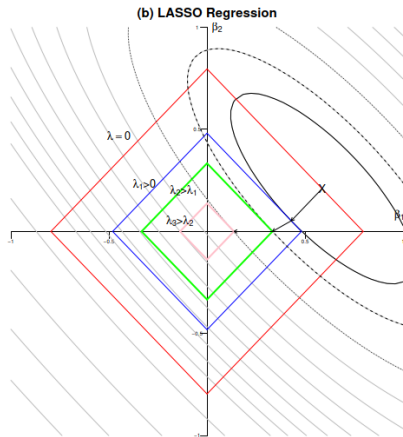
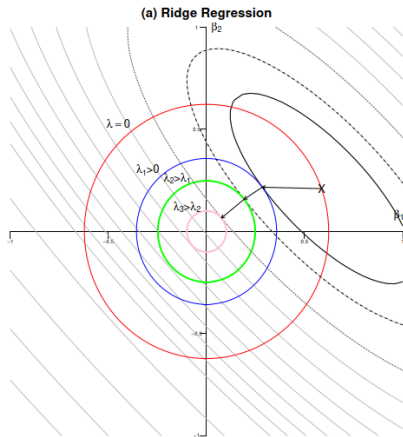
Combining (2.18), (2.20), and (iii), we obtain the LASSO of Tibshirani (1996) for the orthonormal case as

$$\hat{\beta}_j^{\text{LASSO}}(\lambda) = \text{sgn}(\hat{\beta}_j) \left(|\hat{\beta}_j| - \lambda \right)_+, \quad (2.21)$$

where $a_+ = \max(0, a)$ and $\hat{\beta}_j$ is the j th element of the OLS estimator $\hat{\beta}$.

LASSO cont.

Bivariate geometrical comparison, i.e., the case $p = 2$ with $\boldsymbol{\beta} = (\beta_1, \beta_2)^\top$.



Week3-Lab1

Adopted from Efron et al. (2004)³

TABLE 1

Diabetes study: 442 diabetes patients were measured on 10 baseline variables; a prediction model was desired for the response variable, a measure of disease progression one year after baseline

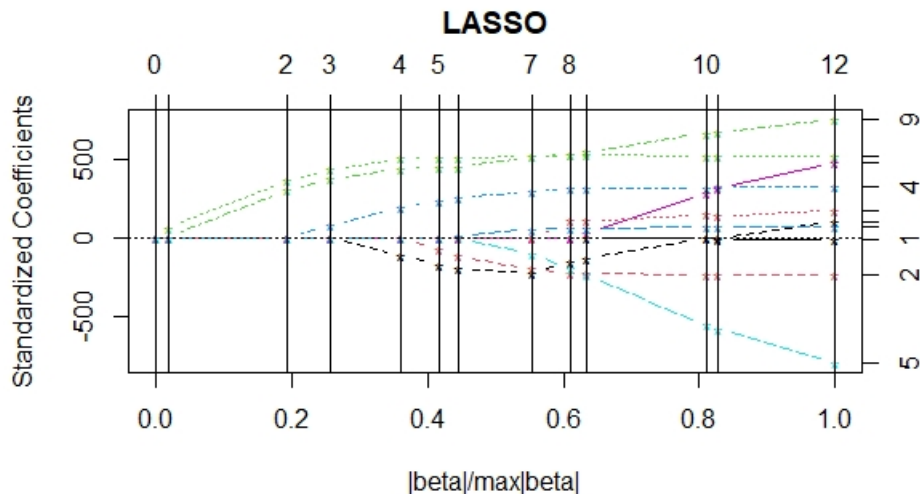
Patient	AGE	SEX	BMI	BP	Serum measurements						Response
	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	x ₇	x ₈	x ₉	x ₁₀	y
1	59	2	32.1	101	157	93.2	38	4	4.9	87	151
2	48	1	21.6	87	183	103.2	70	3	3.9	69	75
3	72	2	30.5	93	156	93.6	41	4	4.7	85	141
4	24	1	25.3	84	198	131.4	40	5	4.9	89	206
5	50	1	23.0	101	192	125.4	52	4	4.3	80	135
6	23	1	22.6	89	139	64.8	61	2	4.2	68	97
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
441	36	1	30.0	95	201	125.2	42	5	5.1	85	220
442	36	1	19.6	71	250	133.2	97	3	4.6	92	57

The response is a quantitative measure of disease progression one year after baseline.

³Efron, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004) Least angle regression, Ann. Statist., 32(2), 407-499.

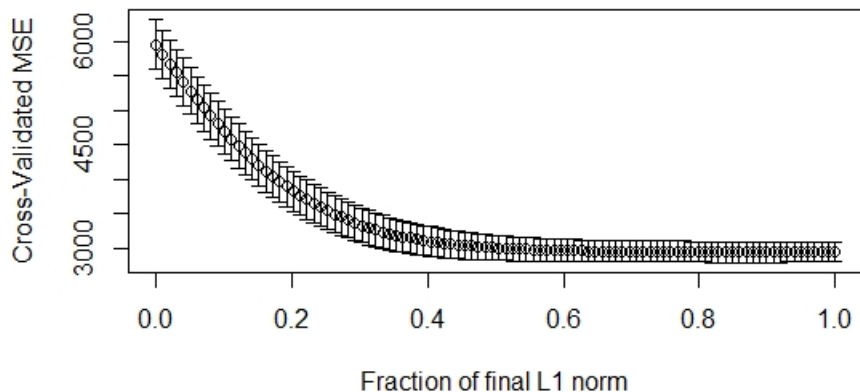
Week3-Lab1 cont.

Analyzing "Diabetes" data with "lars".



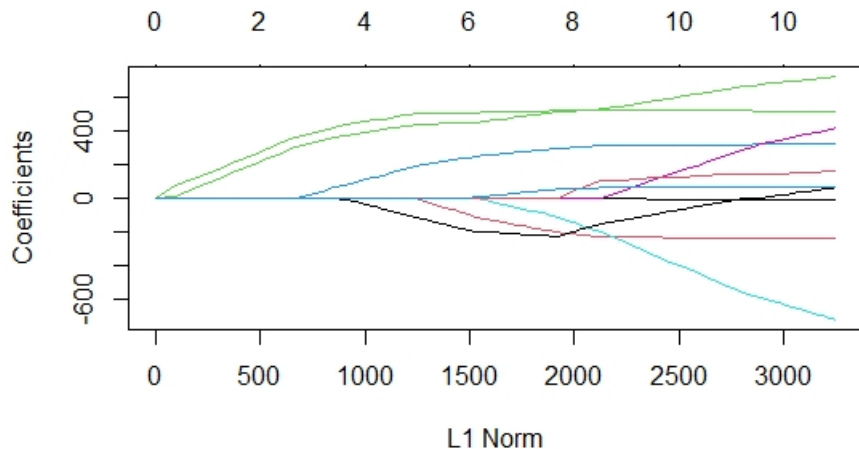
Week3-Lab1 cont.

Cross-validated mean squared prediction error:

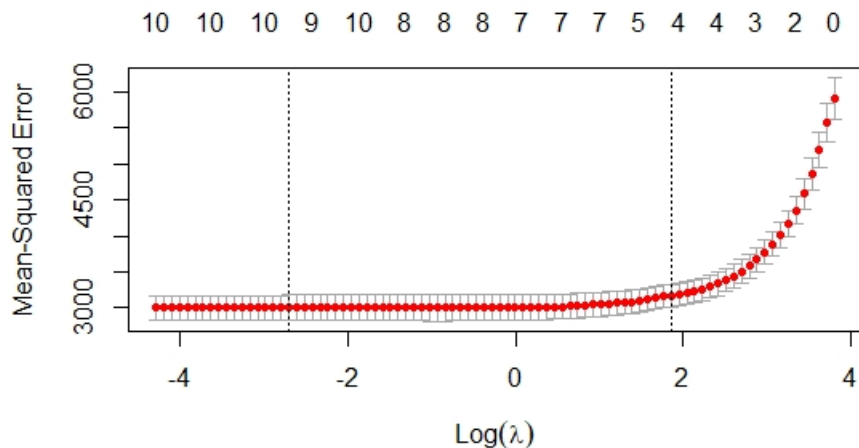


Week3-Lab1 cont.

Package "glmnet"



Week3-Lab1 cont.



Penalty functions

We want $p_\lambda(|\beta|) \geq p_\lambda(|\beta'|)$ when $|\beta| \geq |\beta'|$. This forces the optimization to set most coefficients in β to $\mathbf{0}$, thereby performing selection.

(Convex)Penalty
Bridge

$$p_\lambda(|\beta_i|) \\ \lambda|\beta_i|^\gamma$$

Description

$$\gamma > 0$$

$\gamma < 1$: unbiased

$\gamma \leq 1$: sparse

$\gamma > 1$: continues

$\gamma \geq 1$: convex

$$0 < \alpha < 1$$

β_i^* is a $\sqrt{n}$ -consistent estimator; $\gamma > 0$

Elastic net
Adaptive LASSO

$$p_\lambda(|\beta|) = \lambda (\alpha|\beta_i| + (1 - \alpha)\beta_i^2) \\ \lambda \hat{w}_i |\beta_i|, \hat{w}_i = \frac{1}{|\beta_i^*|^\gamma}$$

Fused LASSO

$$\lambda_1 |\beta_i| + \lambda_2 |\beta_i - \beta_{i-1}|$$

Penalty functions cont.

Concave penalties fulfill all three good properties.

Penalty	$p_\lambda(\beta_i)$	Description
SCAD	$\int_0^{ \beta_i } \min\left\{1, \frac{(a-t/\lambda)_+}{a-1}\right\} dt =$ $\left\{ \begin{array}{ll} \lambda \beta_i & \beta_i < \lambda \\ -\frac{\beta_i^2 - 2a\lambda \beta_i + \lambda^2}{2(a-1)} & \lambda < \beta_i \leq a\lambda \\ \frac{1}{2}(a+1)\lambda^2 & \beta_i > a\lambda \end{array} \right.$	$a > 2$
MCP	$\lambda \int_0^{ \beta_i } \left(1 - \frac{t}{a\lambda}\right)_+ dt$	$a > 1$
Atan	$\lambda \left(\gamma + \frac{2}{\pi}\right) \arctan\left(\frac{ \beta_i }{\gamma}\right)$	$\gamma > 0$
Mnet	$p_{\lambda_1}(\beta_i) + \lambda_2 \beta_i^2$	

Week3-Lab2

Here, we fit the Enet (Elastic net) to a high-dimensional data. Gene expression data (3051 genes and 38 tumor mRNA samples) from the leukemia microarray study

A list with the following elements:

<code>X</code>	a (38 x 3051) matrix giving the expression levels of 3051 genes for 38 leukemia patients. Each row corresponds to a patient, each column to a gene.
<code>Y</code>	a numeric vector of length 38 giving the cancer class of each patient.
<code>gene.names</code>	a matrix containing the names of the 3051 genes for the gene expression matrix <code>X</code> . The three columns correspond to the gene 'index', 'ID', and 'Name', respectively.

Thank you for your attention.

